

Creating Trigonometric Models

You can write the equation of a sinusoidal function if you know its characteristics.

EXAMPLE 1

Write the equation of a function in the form $y = A \sin Bx + C$ or $y = A \cos Bx + C$ for the given characteristics.

- A** frequency = $\frac{4}{\pi}$; amplitude = 2.25; midline: $y = 4.5$;
 y -intercept: $(0, 4.5)$; $A > 0$

SOLUTION

To write a function with the given characteristics, determine the values of A , B , and C .

Since amplitude = 2.25, $|A| = 2.25$. The equation of the midline is $y = 4.5$, so $C = 4.5$. You can use the frequency formula to determine B .

$$\begin{aligned}\text{frequency} &= \frac{B}{2\pi} \\ \frac{4}{\pi} &= \frac{B}{2\pi} \\ B &= 8\end{aligned}$$

Since the y -intercept is on the midline, substitute the values of A , B , and C into the sine function.

$$y = 2.25 \sin(8x) + 4.5$$

B period = 24; absolute minimum = 5; absolute maximum = 12;
 y -intercept: $(0, 12)$; $A > 0$

SOLUTION

Determine the values of A , B , and C .

$$\text{amplitude} = \frac{\text{maximum} - \text{minimum}}{2} = \frac{12 - 5}{2} = 3.5$$

So $|A| = 3.5$.

$$\text{period} = \frac{2\pi}{B}$$

$$24 = \frac{2\pi}{B}$$

$$24B = 2\pi$$

$$B = \frac{\pi}{12}$$

$$\text{equation of midline: } y = \frac{\text{maximum} + \text{minimum}}{2} = \frac{12 + 5}{2} = 8.5$$

So $C = 8.5$.

Since the y -intercept is a maximum, substitute the values of A , B , and C into the cosine function.

$$y = 3.5 \cos\left(\frac{\pi x}{12}\right) + 8.5 \quad \blacksquare$$

EXAMPLE 2

The Singapore Flyer is a 165 m tall Ferris wheel that takes 30 min to make 1 full revolution. Assuming you start at ground level, determine the equation of a sinusoidal function, where your height in meters (y) is a function of time in hours (x).

SOLUTION

Determine the values of A , B , and C . Since the Ferris wheel starts a revolution from ground level, both the minimum and the y -intercept are $(0, 0)$. Since the Ferris wheel is 165 m tall, the maximum is 165.

$$\text{amplitude} = \frac{\text{maximum} - \text{minimum}}{2} = \frac{165 - 0}{2} = 82.5$$

So $|A| = 82.5$.

$$\text{period} = \frac{2\pi}{B}$$

$$0.5 = \frac{2\pi}{B}$$

$$0.5B = 2\pi$$

$$B = 4\pi$$

► **Think About It** The period is 30 min = 0.5 h.

$$\text{equation of midline: } y = \frac{\text{maximum} + \text{minimum}}{2} = \frac{165 + 0}{2} = 82.5$$

So $C = 82.5$.

Since you start at the Ferris wheel's minimum value, a cosine function with $A < 0$ should be used. Substitute the values you found for A , B , and C into the cosine function.

$$y = -82.5 \cos(4\pi x) + 82.5 \quad \blacksquare$$